

TRANSPORT PHENOMENA

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TRANSPORT PHENOMENA

Basic Overview

Over the last few decades, the subject has revolutionized the way Chemical Engineering science is taught. This course deals with the unified treatment of the different transport processes, ubiquitous in industry as well as in nature. Momentum, heat and mass transfer are taught together due to the underlying similarities of the mathematics and molecular mechanisms describing such processes.

Books

1. **Transport Phenomena by Bird, Stewart and Lightfoot (Wiley)**
2. Transport Phenomena Fundamentals by J. L. Plawsky (CRC Press)
3. Heat and Mass Transfer – A Transport Phenomena Approach by K. S. Gandhi (New Age)
4. **Fluid Mechanics by R. W. Fox, P. J. Pritchard and A. T. McDonald (Wiley)**
5. **Introduction to Heat and Mass Transfer by F. P. Incropera & D. P. DeWitt (Wiley)**

Grading: 2-3 class tests for TA, relative grading

Course Plan

Formulation and solution of momentum transfer in laminar flow.

Navier-Stokes equation and its applications

Boundary Layer concepts, boundary layer thicknesses (disturbance, displacement and momentum), Blasius solution for flow over a flat plate

Use of momentum integral equation, turbulent boundary layers, fluid flow about immersed bodies, drag

Formulation and solution of heat transfer in laminar flow

Development and use of energy equation

Transient conduction - lumped capacitance, analytical solutions and other methods.

Formulation and solution of mass transfer in laminar flow. Development and use of species balance equation

Introduction to convective flow, natural convection, relevant examples from heat and mass transfer

Mathematical treatment of the similarities between heat, mass and momentum transfer, similarity parameters, and relevant analogies.

Solution of coupled heat, mass and momentum transfer problems based on analogy.

Phenomenological Relations

Governing Equations

Re

Pr

Sc

f

Nu

Sh

$f = f(\text{Re})$

$\text{Nu} = f(\text{???})$

$\text{Sh} = f(\text{???})$

TRANSPORT PHENOMENA

Application of Navier Stokes Equations

Continuity Equation

$$\frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z} = 0$$

Equation of Motion – Navier Stokes equation

$$\rho D\bar{v} / Dt = -\nabla p - [\nabla \cdot \bar{\tau}] + \rho g$$

Cartesian Coordinate – z component

$$\begin{aligned} & \rho \left(\frac{\partial v_z}{\partial t} + v_x \frac{\partial v_z}{\partial x} + v_y \frac{\partial v_z}{\partial y} + v_z \frac{\partial v_z}{\partial z} \right) \\ &= - \frac{\partial P}{\partial z} + \mu \left[\frac{\partial^2 v_z}{\partial x^2} + \frac{\partial^2 v_z}{\partial y^2} + \frac{\partial^2 v_z}{\partial z^2} \right] + \rho g_z \end{aligned}$$

Relevant Boundary Conditions

No slip at the liquid-solid interface

No shear at the liquid-vapor interface

The Equation of Continuity

$$[\partial\rho/\partial t + (\nabla \cdot \rho\mathbf{v}) = 0]$$

Cartesian coordinates (x, y, z):

$$\frac{\partial\rho}{\partial t} + \frac{\partial}{\partial x}(\rho v_x) + \frac{\partial}{\partial y}(\rho v_y) + \frac{\partial}{\partial z}(\rho v_z) = 0 \quad (\text{B.4-1})$$

Cylindrical coordinates (r, θ , z):

$$\frac{\partial\rho}{\partial t} + \frac{1}{r} \frac{\partial}{\partial r}(\rho r v_r) + \frac{1}{r} \frac{\partial}{\partial \theta}(\rho v_\theta) + \frac{\partial}{\partial z}(\rho v_z) = 0 \quad (\text{B.4-2})$$

Spherical coordinates (r, θ , ϕ):

$$\frac{\partial\rho}{\partial t} + \frac{1}{r^2} \frac{\partial}{\partial r}(\rho r^2 v_r) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta}(\rho v_\theta \sin \theta) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \phi}(\rho v_\phi) = 0 \quad (\text{B.4-3})$$

^a When the fluid is assumed to have constant mass density ρ , the equation simplifies to $(\nabla \cdot \mathbf{v}) = 0$.

The Equation of Motion for a Newtonian Fluid with constant μ , ρ

The Navier Stokes Equation

Cartesian coordinates (x, y, z):

$$\rho \left(\frac{\partial v_x}{\partial t} + v_x \frac{\partial v_x}{\partial x} + v_y \frac{\partial v_x}{\partial y} + v_z \frac{\partial v_x}{\partial z} \right) = -\frac{\partial p}{\partial x} + \mu \left[\frac{\partial^2 v_x}{\partial x^2} + \frac{\partial^2 v_x}{\partial y^2} + \frac{\partial^2 v_x}{\partial z^2} \right] + \rho g_x$$

$$\rho \left(\frac{\partial v_y}{\partial t} + v_x \frac{\partial v_y}{\partial x} + v_y \frac{\partial v_y}{\partial y} + v_z \frac{\partial v_y}{\partial z} \right) = -\frac{\partial p}{\partial y} + \mu \left[\frac{\partial^2 v_y}{\partial x^2} + \frac{\partial^2 v_y}{\partial y^2} + \frac{\partial^2 v_y}{\partial z^2} \right] + \rho g_y$$

$$\rho \left(\frac{\partial v_z}{\partial t} + v_x \frac{\partial v_z}{\partial x} + v_y \frac{\partial v_z}{\partial y} + v_z \frac{\partial v_z}{\partial z} \right) = -\frac{\partial p}{\partial z} + \mu \left[\frac{\partial^2 v_z}{\partial x^2} + \frac{\partial^2 v_z}{\partial y^2} + \frac{\partial^2 v_z}{\partial z^2} \right] + \rho g_z$$

Cylindrical coordinates (r, θ, z):

$$\begin{aligned} \rho \left(\frac{\partial v_r}{\partial t} + v_r \frac{\partial v_r}{\partial r} + \frac{v_\theta}{r} \frac{\partial v_r}{\partial \theta} + v_z \frac{\partial v_r}{\partial z} - \frac{v_\theta^2}{r} \right) &= -\frac{\partial p}{\partial r} + \mu \left[\frac{\partial}{\partial r} \left(\frac{1}{r} \frac{\partial}{\partial r} (r v_r) \right) + \frac{1}{r^2} \frac{\partial^2 v_r}{\partial \theta^2} + \frac{\partial^2 v_r}{\partial z^2} - \frac{2}{r^2} \frac{\partial v_\theta}{\partial \theta} \right] + \rho g_r \\ \rho \left(\frac{\partial v_\theta}{\partial t} + v_r \frac{\partial v_\theta}{\partial r} + \frac{v_\theta}{r} \frac{\partial v_\theta}{\partial \theta} + v_z \frac{\partial v_\theta}{\partial z} + \frac{v_r v_\theta}{r} \right) &= -\frac{1}{r} \frac{\partial p}{\partial \theta} + \mu \left[\frac{\partial}{\partial r} \left(\frac{1}{r} \frac{\partial}{\partial r} (r v_\theta) \right) + \frac{1}{r^2} \frac{\partial^2 v_\theta}{\partial \theta^2} + \frac{\partial^2 v_\theta}{\partial z^2} + \frac{2}{r^2} \frac{\partial v_r}{\partial \theta} \right] + \rho g_\theta \\ \rho \left(\frac{\partial v_z}{\partial t} + v_r \frac{\partial v_z}{\partial r} + \frac{v_\theta}{r} \frac{\partial v_z}{\partial \theta} + v_z \frac{\partial v_z}{\partial z} \right) &= -\frac{\partial p}{\partial z} + \mu \left[\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial v_z}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 v_z}{\partial \theta^2} + \frac{\partial^2 v_z}{\partial z^2} \right] + \rho g_z \end{aligned}$$

Spherical coordinates (r, θ, φ):

$$\begin{aligned} \rho \left(\frac{\partial v_r}{\partial t} + v_r \frac{\partial v_r}{\partial r} + \frac{v_\theta}{r} \frac{\partial v_r}{\partial \theta} + \frac{v_\phi}{r \sin \theta} \frac{\partial v_r}{\partial \phi} - \frac{v_\theta^2 + v_\phi^2}{r} \right) &= -\frac{\partial p}{\partial r} \\ &+ \mu \left[\frac{1}{r^2} \frac{\partial^2}{\partial r^2} (r^2 v_r) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial v_r}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 v_r}{\partial \phi^2} \right] + \rho g_r \\ \rho \left(\frac{\partial v_\theta}{\partial t} + v_r \frac{\partial v_\theta}{\partial r} + \frac{v_\theta}{r} \frac{\partial v_\theta}{\partial \theta} + \frac{v_\phi}{r \sin \theta} \frac{\partial v_\theta}{\partial \phi} + \frac{v_r v_\theta - v_\phi^2 \cot \theta}{r} \right) &= -\frac{1}{r} \frac{\partial p}{\partial \theta} \\ &+ \mu \left[\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial v_\theta}{\partial r} \right) + \frac{1}{r^2} \frac{\partial}{\partial \theta} \left(\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} (v_\theta \sin \theta) \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 v_\theta}{\partial \phi^2} + \frac{2}{r^2} \frac{\partial v_r}{\partial \theta} - \frac{2 \cot \theta}{r^2 \sin \theta} \frac{\partial v_\phi}{\partial \phi} \right] + \rho g_\theta \\ \rho \left(\frac{\partial v_\phi}{\partial t} + v_r \frac{\partial v_\phi}{\partial r} + \frac{v_\theta}{r} \frac{\partial v_\phi}{\partial \theta} + \frac{v_\phi}{r \sin \theta} \frac{\partial v_\phi}{\partial \phi} + \frac{v_\phi v_r + v_\theta v_\phi \cot \theta}{r} \right) &= -\frac{1}{r \sin \theta} \frac{\partial p}{\partial \phi} \\ &+ \mu \left[\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial v_\phi}{\partial r} \right) + \frac{1}{r^2} \frac{\partial}{\partial \theta} \left(\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} (v_\phi \sin \theta) \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 v_\phi}{\partial \phi^2} + \frac{2}{r^2 \sin \theta} \frac{\partial v_r}{\partial \phi} + \frac{2 \cot \theta}{r^2 \sin \theta} \frac{\partial v_\theta}{\partial \phi} \right] + \rho g_\phi \end{aligned}$$

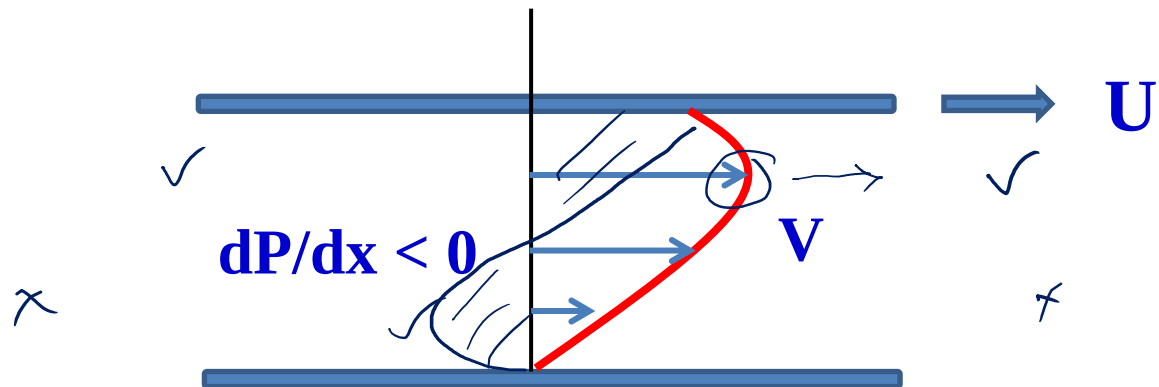
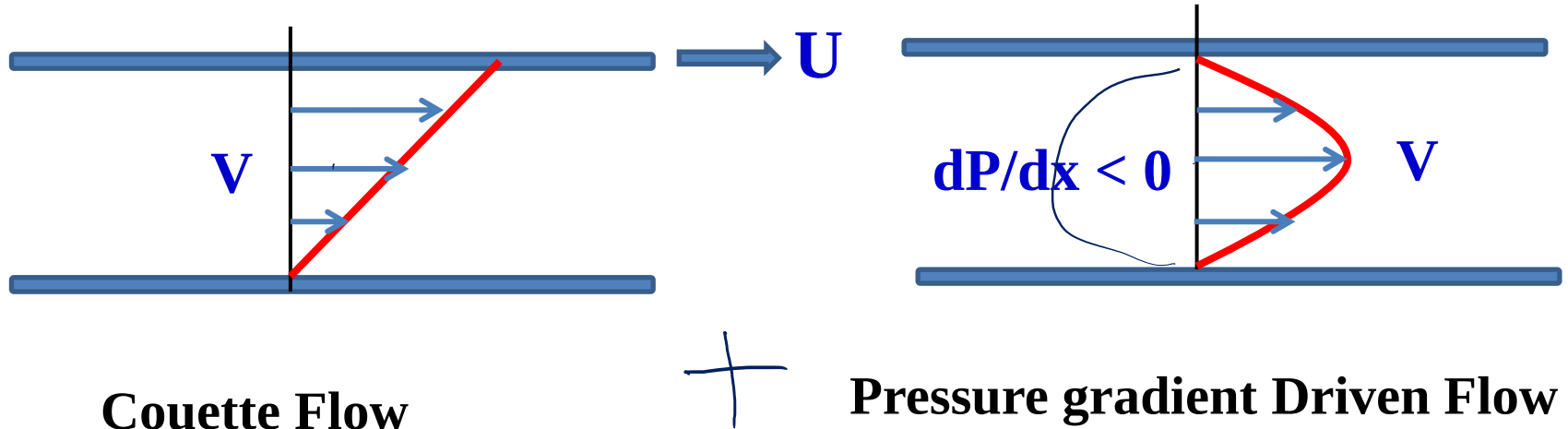
Transport Phenomena - Momentum Transfer

The record-write head for a computer disk storage system floats above the spinning disk on a very thin film of air (the film thickness is 0.5micron). The head location is 150mm from the disk centre line, the disk spins at 3600rpm. The record-write head is 10mm x 10mm square. Determine for standard air (density= 1.23kg/m^3 , viscosity= $1.78 \times 10^{-5} \text{ kg/(m.s)}$) in the gap between the head and the disk -

(a) the Reynold's number of the flow, (b) the viscous shear stress and (c) the power required to overcome viscous shear stress.

For such a small gap the flow can be considered as flow between parallel plates.

Flow between parallel plates



Couette and Pressure gradient Driven Flow

Solution

Since the gap is too small, the situation can be approximated by the case of flow between two parallel plates with one plate moving and the other stationary with no applied pressure gradient.

Here,

$$V = R\omega = 0.15 \text{ m} \times 3600 \text{ rev/min} \times 2 \pi \text{ rad/rev} \times 1/60 \text{ min/s} = 56.5 \text{ m/s}$$

$$\text{Re} = (\rho V a)/\mu = (V a)/(\mu/\rho) = 56.5 \text{ m/s} \times 0.5 \times 10^{-6} \text{ m} / (1.45 \times 10^{-5} \text{ m}^2/\text{s}) = 1.95 \text{ (Laminar Flow)}$$

For Couette flow between two parallel plates, with no applied pressure gradient, the velocity profile will be linear with no-slip boundary conditions at the bottom (stationary) and top (moving) plates.

Therefore, the shear stress, τ , can be expressed as

$$\tau = \mu du/dy = \mu V/a \text{ (linear velocity profile)}$$

$$\tau = 1.78 \times 10^{-5} \text{ kg / (m.s) } \times 56.5 \text{ m/s } \times 1/(0.5 \times 10^{-6} \text{ m}) = 2.01 \text{ KN/m}^2$$

Therefore,

$$\text{Force, } F = \tau A = \tau \times (W \times L), \text{ Torque, } T = F \times R = \tau \times (WL) \times R$$

$$\text{Power dissipation rate. } P = T \times \omega = \tau WL R \omega$$

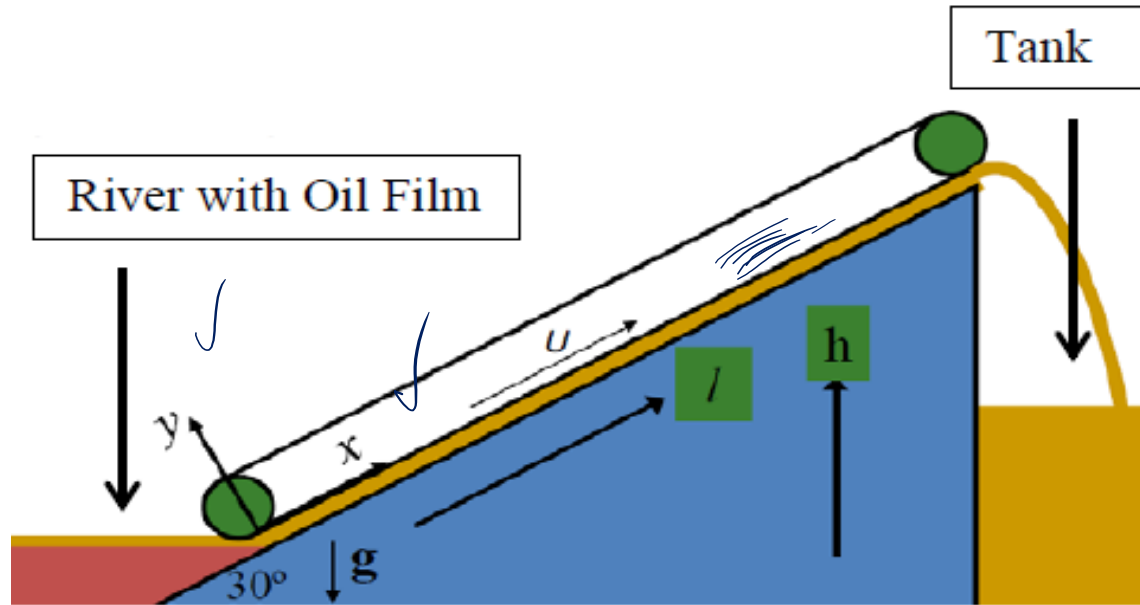
$$P = 2.01 \times 10^{-3} \times 0.01 \times 0.01 \times 0.15 \times 3600 \times 2\pi \times 1/60 = 11.4 \text{ W}$$

Thus the power required to overcome viscous shear stress is 11.4 W

Transport Phenomena - Momentum Transfer

Problem 2

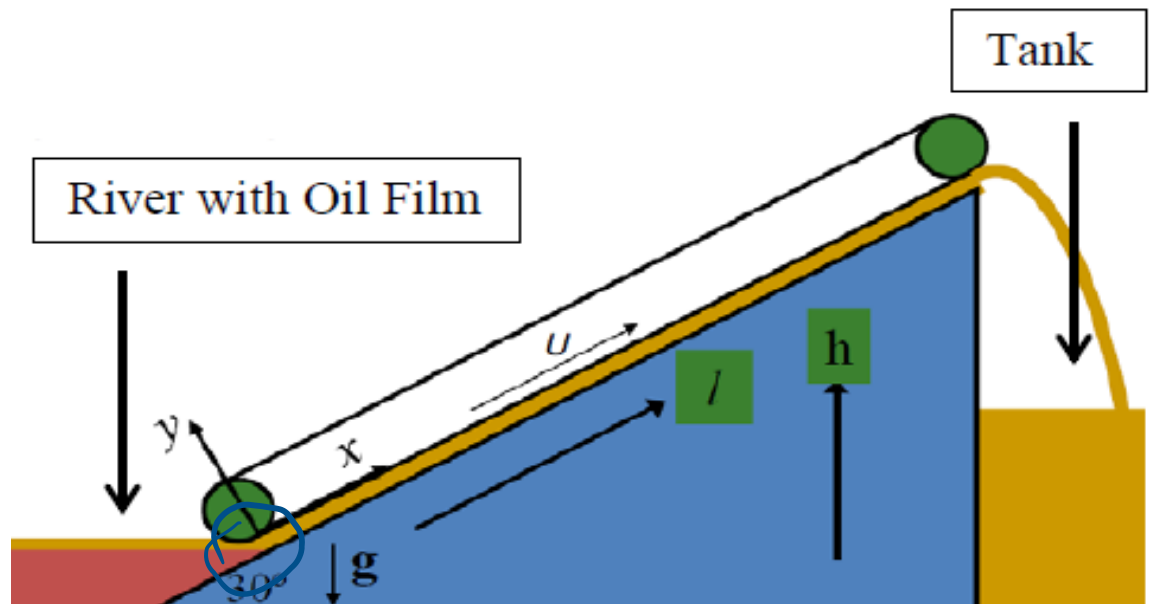
An oil skimmer uses a 5 m wide x 6 m long moving belt above a fixed platform ($\theta = 30^\circ$) to skim oil off of rivers ($T = 10^\circ\text{C}$). The belt travels at 3 m/s. The distance between the belt and the fixed platform is 2 mm.



The belt discharges into an open tank on the ship. The fluid is actually a mixture of oil and water. To simplify the analysis, assume crude oil dominates. Find the discharge of oil into the tank on the ship, the force acting on the belt and the power required (kW) to move the belt. For oil: $\rho = 860 \text{ kg/m}^3$, viscosity, $\mu = 1 \times 10^{-2} \text{ N.s/m}^2$

The x-component of the equation of motion is given as

$$\rho \left(\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + w \frac{\partial u}{\partial z} \right) = - \frac{\partial P}{\partial x} + \mu \left[\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right] + \rho g_x$$



Considering one-dimensional flow ($v = 0 = w$), no applied pressure gradient, u is a function of only y , and a steady state process, the above equation reduces to the following governing equation

$$\frac{d^2u}{dy^2} = \frac{\rho g}{\mu} \sin\theta$$

The relevant boundary conditions (no slip) are

$$\text{BC 1 At } y = 0 \quad u = 0$$

$$\text{BC 2 At } y = h, \quad u = U$$

Therefore,

$$\frac{du}{dy} = \frac{\rho g \sin \theta}{\mu} y + A \quad \text{and} \quad u = \left(\frac{\rho g \sin \theta}{\mu} \right) \frac{y^2}{2} + Ay + B$$

Applying the BC s $\rightarrow$ $B = 0$ and $A = \left(\frac{\rho g \sin \theta}{\mu} \right) \frac{h}{2} + \frac{u}{h}$

Thus

$$u = - \left(\frac{\rho g \sin \theta}{\mu} \right) \left(\frac{hy}{2} - \frac{y^2}{2} \right) + \frac{Uy}{h}$$

The volumetric flow rate per unit width of the film is given by

$$Q = \int_0^h u dy = - \int_0^h \left[\left(\frac{\rho g \sin \theta}{\mu} \right) \left(\frac{hy}{2} - \frac{y^2}{2} \right) + \frac{Uy}{h} \right] dy = - \frac{\rho g h^3}{12\mu} \sin \theta + \frac{Uh}{2}$$

$$Q = - \frac{860 \times 9,81 \times 0.002}{12 \times 10^{-2}} \sin 30 + \frac{3 \times 0.002}{2} = 0.0027 \frac{m^2}{s} \text{ (per unit width)}$$

For the width of 5 m, the volumetric flow rate will be equal to 0.0135 m³/s.

Evaluate $\tau = \mu \, du/dy$ at the moving belt from the expression for velocity as

$$\frac{du}{dy} = - \left(\frac{\rho g \sin \theta}{\mu} \right) \left(\frac{h}{2} - y \right) + \frac{U}{h}$$

And at the moving belt,

$$\tau = \mu \left(\frac{du}{dy} \right)_{y=h} = \left(\frac{\rho g \sin \theta}{2} \right) h + \frac{\mu U}{h}$$



Using the values $h = 0.002m$, $\mu = 10^{-2}N.s/m^2$ and $U = 3 \text{ m/s}$

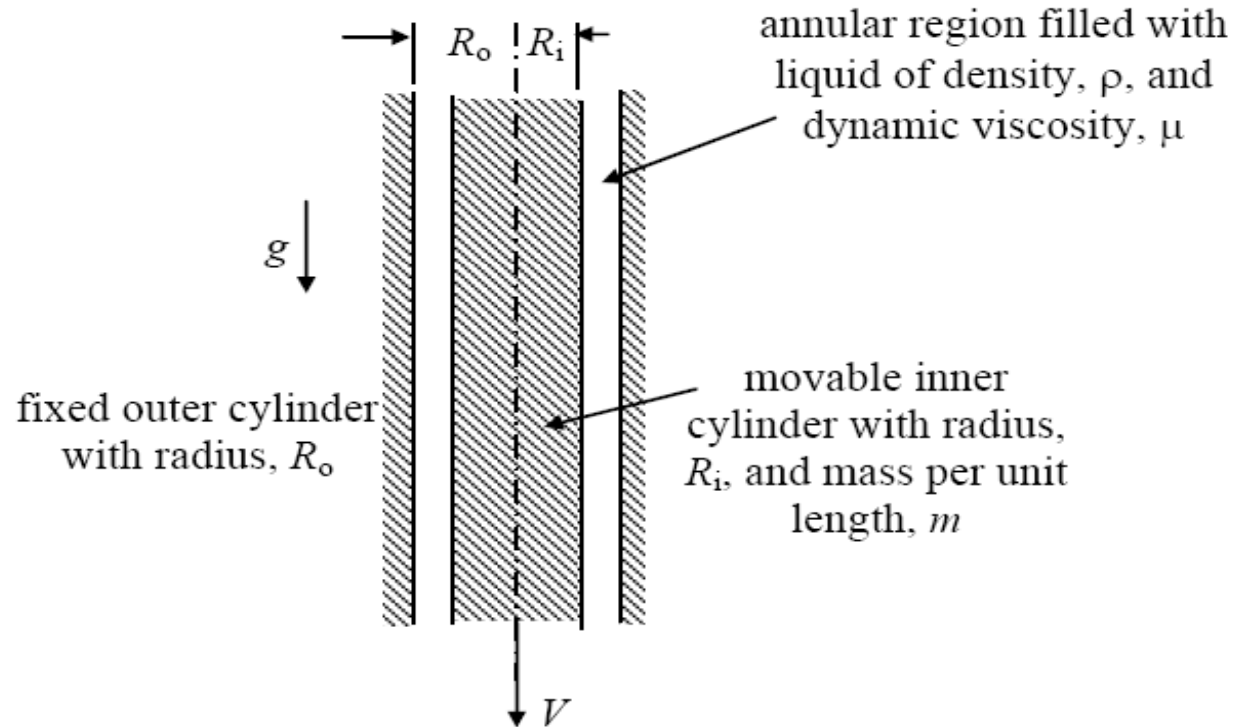
$$\tau \Big|_{\text{at the belt}} = 19.21 \text{ N/m}^2$$

$$\text{Power} = (\tau \times L \times W) U = 19.21 \times 6 \times 5 \times 3 = 1.73 \text{ KW}$$

Transport Phenomena - Momentum Transfer

Problem 3

Consider two concentric cylinders with a Newtonian liquid of constant density, ρ , and viscosity, μ , contained between them. The outer pipe, with radius, R_o , is fixed while the inner pipe, with radius, R_i , and mass per unit length, m , falls under the action of



gravity at a constant speed. There is no pressure gradient within the flow and no swirl velocity component. Determine the vertical speed, V , of the inner cylinder as a function of the following parameters: g , R_o , R_i , m , ρ , and μ . The space between the two cylinders is **not** 'too small' compared to the radii of the cylinders

Solution

For the inner cylinder moving at constant velocity, the downward force is exactly balanced by the viscous force as

$$(\tau_w A_w) \big|_{Inner\ Cylinder} = m L g$$

The z component of the equation of motion in cylindrical coordinate is

$$\begin{aligned} & \rho \left(\frac{\partial v_z}{\partial t} + v_r \frac{\partial v_z}{\partial r} + \frac{v_\theta}{r} \frac{\partial v_z}{\partial \theta} + v_z \frac{\partial v_z}{\partial z} \right) \\ &= -\frac{\partial p}{\partial z} + \mu \left[\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial v_z}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 v_z}{\partial \theta^2} + \frac{\partial^2 v_z}{\partial z^2} \right] + \rho g_z \end{aligned}$$

Cancelling the terms with the observations i) steady state, ii) v_z is a function of r only, not of z or θ , iii) no applied pressure gradient

The following simplified form of the NS equation can be obtained.

Governing equation

$$\frac{1}{r} \frac{d}{dr} \left(r \frac{dv_z}{dr} \right) = - \frac{\rho g}{\mu}$$

Upon integration (with C_1 and C_2 being the constants of integration)

$$v_z = - \frac{\rho g r^2}{4\mu} + C_1 \ln r + C_2$$

Boundary Conditions

$$v_z = V \text{ at } r = R_i \quad \Rightarrow \quad V = - \frac{\rho g}{4\mu} R_i^2 + C_1 \ln R_i + C_2$$

$$v_z = 0 \text{ at } r = R_o \quad \Rightarrow \quad 0 = - \frac{\rho g}{4\mu} R_o^2 + C_1 \ln R_o + C_2$$

Therefore
$$V = \frac{\rho g}{4\mu} (R_o^2 - R_i^2) + C_1 \ln \frac{R_i}{R_o}$$

$$C_1 = \frac{1}{\ln \frac{R_i}{R_o}} \left[V - \frac{\rho g}{4\mu} (R_o^2 - R_i^2) \right]$$

$$\frac{dv_z}{dr} = -\frac{\rho g r}{2\mu} + \frac{C_1}{r} \quad \tau_{rz} = \mu \frac{dv_z}{dr} = -\frac{\rho g r}{2} + \frac{C_1 \mu}{r}$$

Since the force on the inner cylinder = force due to gravity

$$\tau \big|_{r=R_i} 2\pi R_i L = m L g$$

Upon substitution for the expression of the shear stress

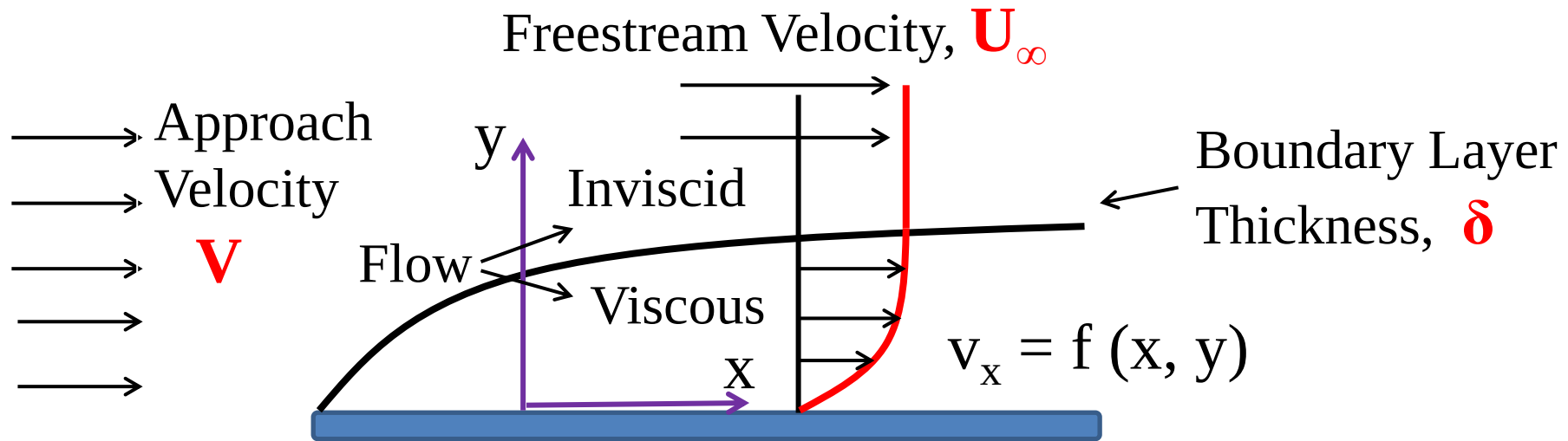
$$V = R_i \ln \frac{R_i}{R_o} \left(\frac{\rho g R_i}{2\mu} - \frac{m g}{2\pi R_i \mu} \right) - \frac{\rho g}{4\mu} (R_i^2 - R_o^2)$$

Order of magnitude analysis of NS Equation

Is it possible to identify the relative magnitudes of the different terms (even approximately)?

It may then be possible to neglect the term(s) that may not play a crucial role in the transport process thereby simplifying NS equations.

External Incompressible Viscous Flow – Boundary Layer



Flow over a flat plate

$v_x = f(x, y)$ Viscous 2D flow inside BL

$v_x = U_\infty$ Inviscid flow outside BL

$v_x = 0.99U_\infty$
at $y = \delta$

δ - boundary layer thickness

Flow Inside the Boundary Layer

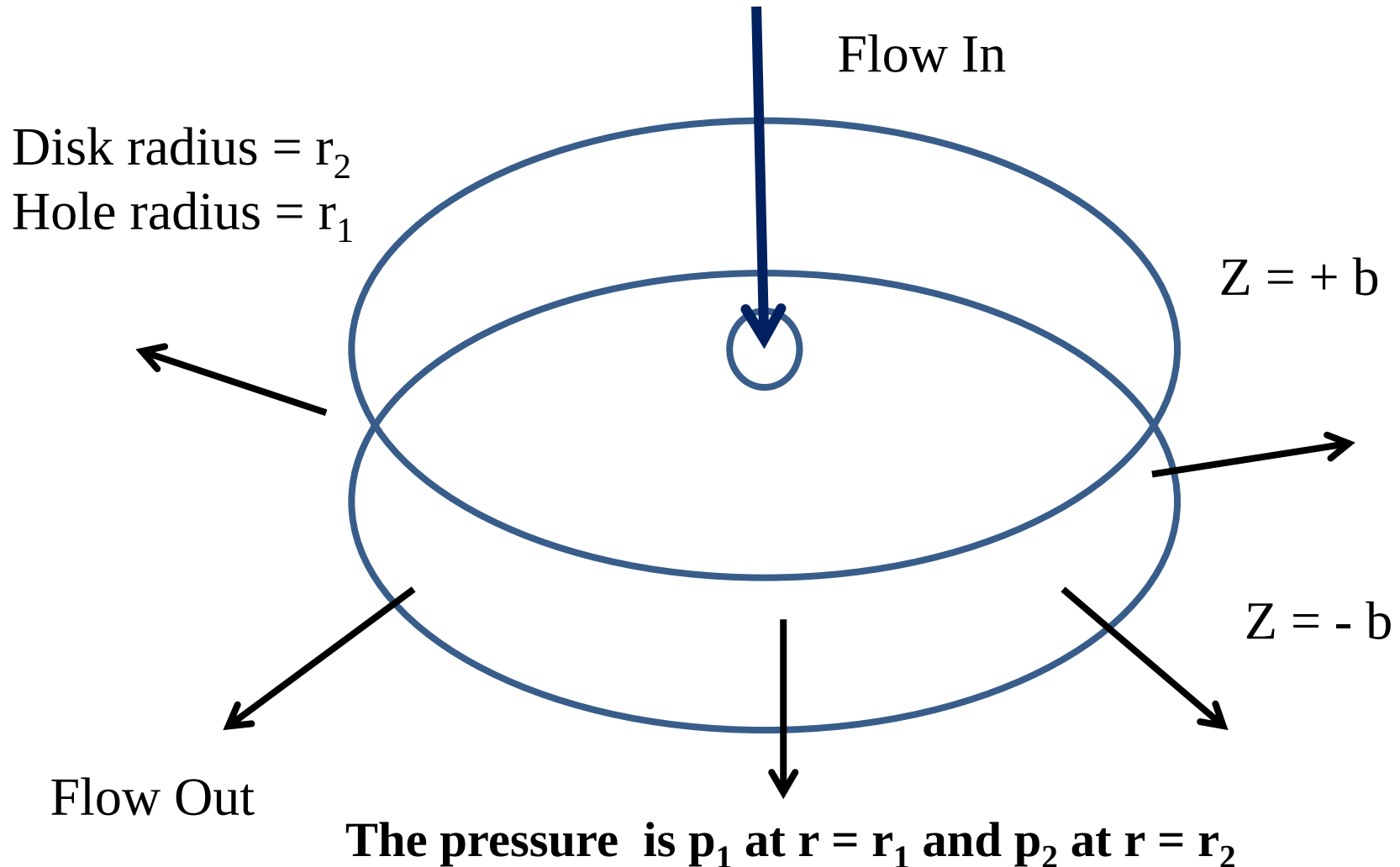
$$v_x \frac{\partial v_x}{\partial x} + v_y \frac{\partial v_x}{\partial y} = \gamma \left(\frac{\partial^2 v_x}{\partial x^2} + \frac{\partial^2 v_x}{\partial y^2} \right)$$

Boundary Layer Approximations

$$v_x \gg v_y \quad \frac{\partial v_x}{\partial y} \gg \frac{\partial v_x}{\partial x} \quad \frac{\partial^2 v_x}{\partial y^2} \gg \frac{\partial^2 v_x}{\partial x^2}$$

$$v_x \frac{\partial v_x}{\partial x} + v_y \frac{\partial v_x}{\partial y} = \gamma \frac{\partial^2 v_x}{\partial y^2}$$

Flow between two parallel disks with liquid entry through a small hole at the centre of the top plate



$$\frac{\partial \rho}{\partial t} + \frac{1}{r} \frac{\partial}{\partial r}(\rho r v_r) + \frac{1}{r} \frac{\partial}{\partial \theta}(\rho v_\theta) + \frac{\partial}{\partial z}(\rho v_z) = 0$$

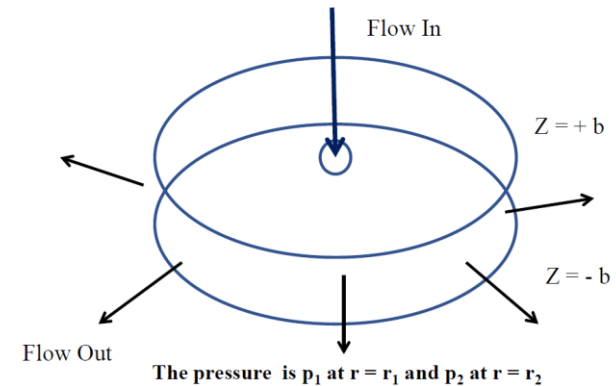
$$\frac{1}{r} \frac{\partial}{\partial r}(r v_r) = 0$$

$$r v_r = \phi \quad v_r$$

$$\Rightarrow v_r = \frac{\phi}{r}, \quad = f(r, z)$$

$$\rho \left(\cancel{\frac{\partial v_r}{\partial t}} + v_r \frac{\partial v_r}{\partial r} + \cancel{\frac{v_\theta}{r} \frac{\partial v_r}{\partial \theta}} - \cancel{\frac{v_\theta^2}{r}} + \cancel{v_z \frac{\partial v_r}{\partial z}} \right)$$

$$= -\frac{\partial P}{\partial r} + \mu \left[\frac{\partial}{\partial r} \left(\frac{1}{r} \frac{\partial}{\partial r} (r v_r) \right) + \frac{1}{r^2} \frac{\partial^2 v_r}{\partial \theta^2} - \frac{2}{r^2} \frac{\partial v_\theta}{\partial \theta} + \frac{\partial^2 v_r}{\partial z^2} \right] + \cancel{\rho g_r}$$



$$\phi = f(z)$$

The governing equation

$$\rho \cancel{v_r} \frac{\partial v_r}{\partial r} = - \frac{\partial P}{\partial r} + \mu \frac{\partial^2 v_r}{\partial z^2} \quad v_r = \frac{\phi(z)}{r}$$

≈ 0

Major assumption on the nature of the flow

CREeping FLOW (CONVECTION
NOT IMP

$$\mu \frac{d^2}{dz^2} (v_r) = - \frac{dP}{dr}$$

$$P = f(r, z)$$

Assume a constant applied pressure difference

$$\frac{\mu}{r} \frac{\partial^2 \phi}{\partial z^2} = \frac{dp}{dr}$$

$$\checkmark \phi = - \frac{\Delta P z^2}{2\mu \ln r_2/r_1} + C_1 z + C_2$$

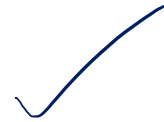
BC $\rightarrow$ NO SLIP $\checkmark$

$$v_r(r, z) = \frac{\phi}{r} = \frac{\Delta P b^2}{2\mu r \ln \frac{r_2}{r_1}} \left[1 - \left(\frac{z}{b} \right)^2 \right]$$

Boundary Conditions

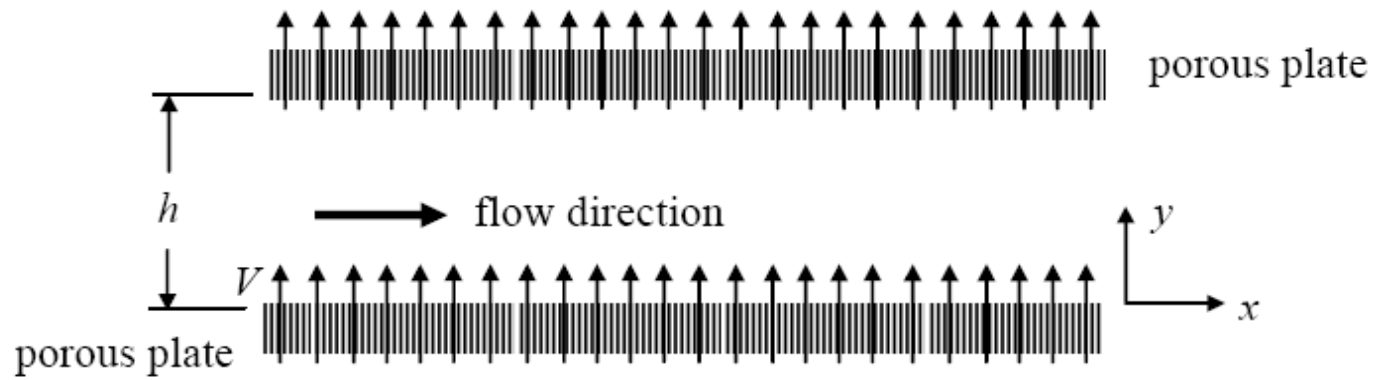


$$v_r(r, z) = \frac{\Delta P b^2}{2 \mu r \ln \frac{r_2}{r_1}} \left[1 - \left(\frac{z}{b} \right)^2 \right]$$



$$Q = 2 \pi \int_{-b}^{+b} r v_r dz = 2 \pi \int_{-b}^{+b} \phi(z) dz = \frac{4 \pi \Delta P b^3}{3 \mu \ln \frac{r_2}{r_1}}$$

An incompressible fluid flows between two porous, parallel flat plates as shown in the figure. An identical fluid is injected at a constant speed V through the bottom plate and simultaneously extracted from the upper plate at the same velocity. Assume the flow to be steady, fully-developed, the pressure gradient in the x -direction is a constant, and neglect body forces.



Determine expressions for the y component of velocity.
Show that the x component of velocity can be expressed as

$$u_x = \frac{h}{\rho V} \left[\frac{\partial p}{\partial x} \right] \left[\left\{ \frac{1 - \exp\left(\frac{\rho V y}{\mu}\right)}{1 - \exp\left(\frac{\rho V h}{\mu}\right)} \right\} - \frac{y}{h} \right]$$

The equation of continuity for fully developed, steady flow in x-direction

$$\cancel{\frac{\partial v_x}{\partial x}} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z} = 0$$

$$= 0 \text{ (Fully dev. flow)}$$

$$\therefore \frac{\partial v_y}{\partial y} = 0 \Rightarrow v = V, \text{ Everywhere in the flow field}$$

The x component of the NS equation

$$\rho \left(\cancel{\frac{\partial v_x}{\partial t}} + v_x \cancel{\frac{\partial v_x}{\partial x}} + v_y \frac{\partial v_x}{\partial y} + \cancel{v_z \frac{\partial v_x}{\partial z}} \right) = - \frac{\partial P}{\partial x} + \mu \left[\cancel{\frac{\partial^2 v_x}{\partial x^2}} + \frac{\partial^2 v_x}{\partial y^2} + \cancel{\frac{\partial^2 v_x}{\partial z^2}} \right] + \cancel{\rho g_x}$$

The governing equation

$$\rho v_y \frac{\partial v_x}{\partial y} = - \frac{\partial P}{\partial x} + \mu \frac{\partial^2 v_x}{\partial y^2}$$

Assumptions:

$\frac{\partial P}{\partial x} = A$

Constant pressure gradient in the x direction ✓

Boundary Conditions

$$v_x = 0 \quad \text{at} \quad y = 0$$

$$v_x = 0 \quad \text{at} \quad y = h$$

$$\mu \frac{d^2 u}{dy^2} = f \vee \frac{du}{dy} - A$$

$\uparrow$
 PR GRAD

$$u = f(\textcircled{0} y)$$

BC

NO SLIP

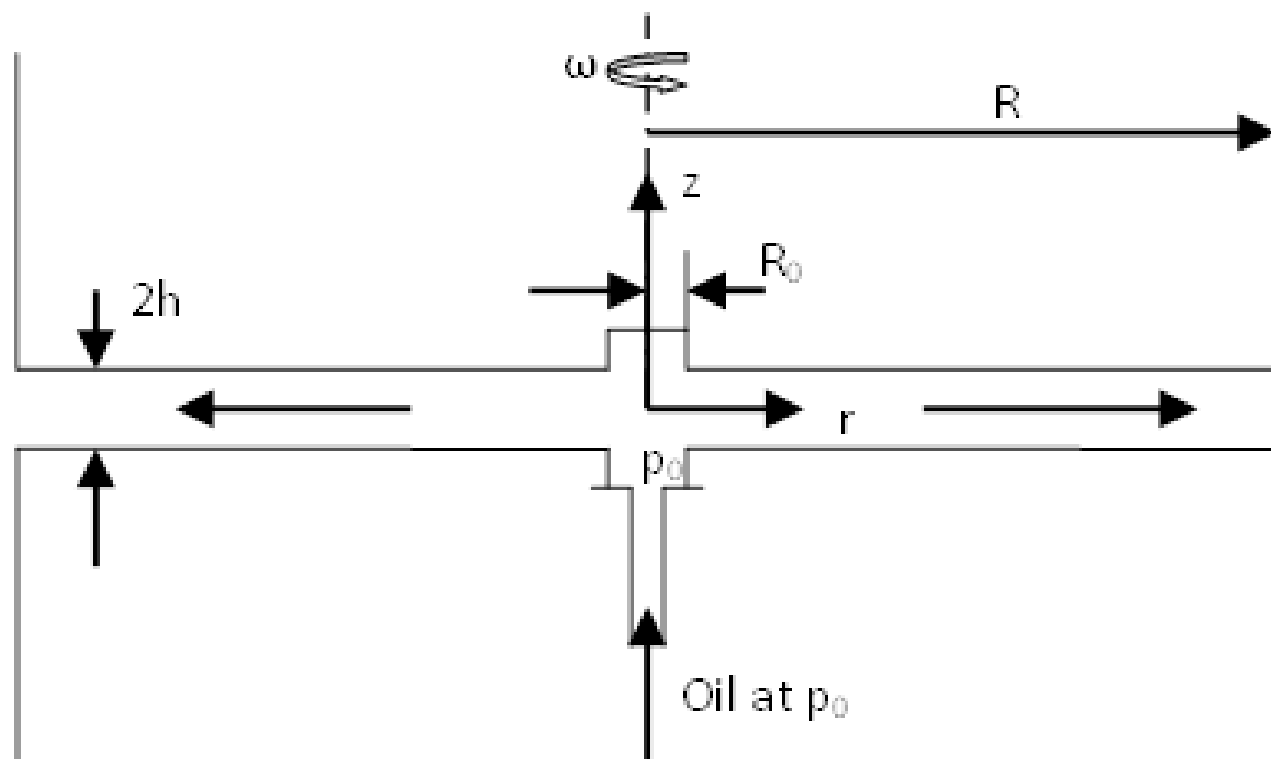
AT $y=0$

$y=h$

Solve the gov. eqnⁿ with
the B.C.s to obtain
The profile as provided
in The Question

A thrust bearing as shown in the figure is lubricated by pumping oil at a high pressure of p_0 . The angular velocity is equal to ω . Note that under laminar conditions, both V_r and V_θ in the thin gap will be non-zero and p is a function of r only. Neglect convective and body force terms in equations of motion. You may also assume that the pressure at $r = R$ is equal to p_{atm} , whereas the pressure from $r = 0$ till $r = R_0$ is equal to p_0 .

- (i) Start with the equation of continuity to obtain the functional form of V_r .
- (ii) Show that $V_\theta = \frac{\omega r}{2} \left(1 + \frac{z}{h} \right)$ satisfies the θ component of the NS equation.
- (iii) Write the r component of NS equation to show that $r \frac{\partial p}{\partial r}$ is a constant.
- (iv) Evaluate the pressure distribution using the boundary conditions
- (v) Find the vertical load the bearing can support and the flow rate of oil required.



- (i) Start with the equation of continuity to obtain the functional form of V_r .

The Equation of Continuity

Cylindrical coordinates (r, θ, z) :

$$\cancel{\frac{\partial \rho}{\partial t}} + \frac{1}{r} \frac{\partial}{\partial r} (\rho r v_r) + \cancel{\frac{1}{r} \frac{\partial}{\partial \theta} (\rho v_\theta)} + \cancel{\frac{\partial}{\partial z} (\rho v_z)} = 0$$

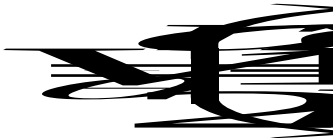
$$\frac{\partial}{\partial r} (r v_r) = 0$$

$$v_r = \frac{f(z)}{r}$$

$$r v_r = f(z)$$

only r

$$\frac{\partial}{\partial \theta} (\rho v_\theta)$$

(ii) Show that  satisfies the θ component of the NS equation.

$$\rho \left(\frac{\partial v_\theta}{\partial t} + v_r \frac{\partial v_\theta}{\partial r} + \frac{v_\theta}{r} \frac{\partial v_\theta}{\partial \theta} + v_z \frac{\partial v_\theta}{\partial z} + \frac{v_r v_\theta}{r} \right) = -\frac{1}{r} \frac{\partial p}{\partial \theta} + \mu \left[\frac{\partial}{\partial r} \left(\frac{1}{r} \frac{\partial}{\partial r} (r v_\theta) \right) + \frac{1}{r^2} \frac{\partial^2 v_\theta}{\partial \theta^2} + \frac{\partial^2 v_\theta}{\partial z^2} + \frac{2}{r^2} \frac{\partial v_r}{\partial \theta} \right] + \rho g_\theta$$

$$0 = -\frac{1}{r} \frac{\partial p}{\partial \theta} + \mu \left[\frac{\partial}{\partial r} \left(\frac{1}{r} \frac{\partial}{\partial r} (r v_\theta) \right) + \frac{1}{r^2} \frac{\partial^2 v_\theta}{\partial \theta^2} + \frac{\partial^2 v_\theta}{\partial z^2} + \frac{2}{r^2} \frac{\partial v_r}{\partial \theta} \right]$$

$$\frac{\partial^2 v_\theta}{\partial z^2} = \frac{\partial}{\partial r} \left(\frac{1}{r} \frac{\partial}{\partial r} (r v_\theta) \right)$$

(iii) Write the r component of NS equation to show that $r \frac{\partial p}{\partial r}$ is a constant

$$\rho \left(\frac{\partial v_r}{\partial t} + v_r \frac{\partial v_r}{\partial r} + \frac{v_\theta}{r} \frac{\partial v_r}{\partial \theta} + v_z \frac{\partial v_r}{\partial z} - \frac{v_\theta^2}{r} \right) = -\frac{\partial p}{\partial r} + \mu \left[\frac{\partial}{\partial r} \left(\frac{1}{r} \frac{\partial}{\partial r} (r v_r) \right) + \frac{1}{r^2} \frac{\partial^2 v_r}{\partial \theta^2} + \frac{\partial^2 v_r}{\partial z^2} - \frac{2}{r^2} \frac{\partial v_\theta}{\partial \theta} \right] + \rho g_r$$

$$0 = -\frac{\partial p}{\partial r} + \mu \left[\frac{\partial}{\partial r} \left(\frac{1}{r} \frac{\partial}{\partial r} (r v_r) \right) + \frac{1}{r^2} \frac{\partial^2 v_r}{\partial \theta^2} + \frac{\partial^2 v_r}{\partial z^2} - \frac{2}{r^2} \frac{\partial v_\theta}{\partial \theta} \right] + \rho g_r$$

$$0 = -\frac{\partial p}{\partial r} + \mu \frac{\partial^2 v_r}{\partial z^2}$$

$$r \frac{\partial p}{\partial r} = \mu \frac{\partial^2 (r v_r)}{\partial z^2}$$

$$r \frac{\partial p}{\partial r} = \mu \frac{\partial^2}{\partial z^2} (r v_r)$$

LHS IS A FUNCTION OF r ONLY

RHS IS A FN of z only

$$\therefore r \frac{\partial p}{\partial r} = \mu \frac{\partial^2}{\partial z^2} (r v_r) = \text{CONST.}, K$$

(iv) Evaluate the pressure distribution using the boundary conditions

$$r \frac{\partial p}{\partial r} = K (\text{CONST})$$

$$\text{At } r = r_0 \quad p = p_0, \quad r = R, \quad p = p_{\text{atm}}$$

$$\Rightarrow K = \frac{p_0 - p_{\text{atm}}}{\ln R_0 / R}$$

$$\Rightarrow p = K \ln r + C$$

∴ Non-linear variation of p with r
⇒ p gradient is not const.

$$\cancel{\partial_z^2 v_r} \frac{\partial^2 (r v_r)}{\partial z^2} = \kappa / \mu$$

$$\text{BC } z = h \quad v_r = 0, \quad z = -h, \\ v_r = 0.$$

$$\Rightarrow v_r = \frac{\kappa}{2\mu r} (z^2 - h^2)$$

$$Q = \int_{-h}^{+h} v_r 2\pi r dz = -\frac{\pi \kappa}{\mu} \frac{4}{3} h^3$$

$$\dot{Q} = \frac{p_0 - p_{atm}}{\ln R/R_0} \frac{4}{3} \frac{\pi h^3}{\mu}$$

Axial load

$$= (p_0 - p_{atm}) \pi R_0^2 + \int_{R_0}^R (p - p_{atm}) 2\pi r dr$$

As

$$p - p_{atm} = k \ln r/R$$

- (v) Find the vertical load the bearing can support and the flow rate of oil required.

$$\text{Load} = \frac{\pi (p_o - p_{\text{atm}}) (R^2 - R_o^2)}{2 \ln R/R_o}$$

$$Q = \frac{p_o - p_{\text{atm}}}{\ln R/R_o} \frac{4}{3} \frac{\pi h^3}{\mu}$$